

Heat Energy

Junior Science

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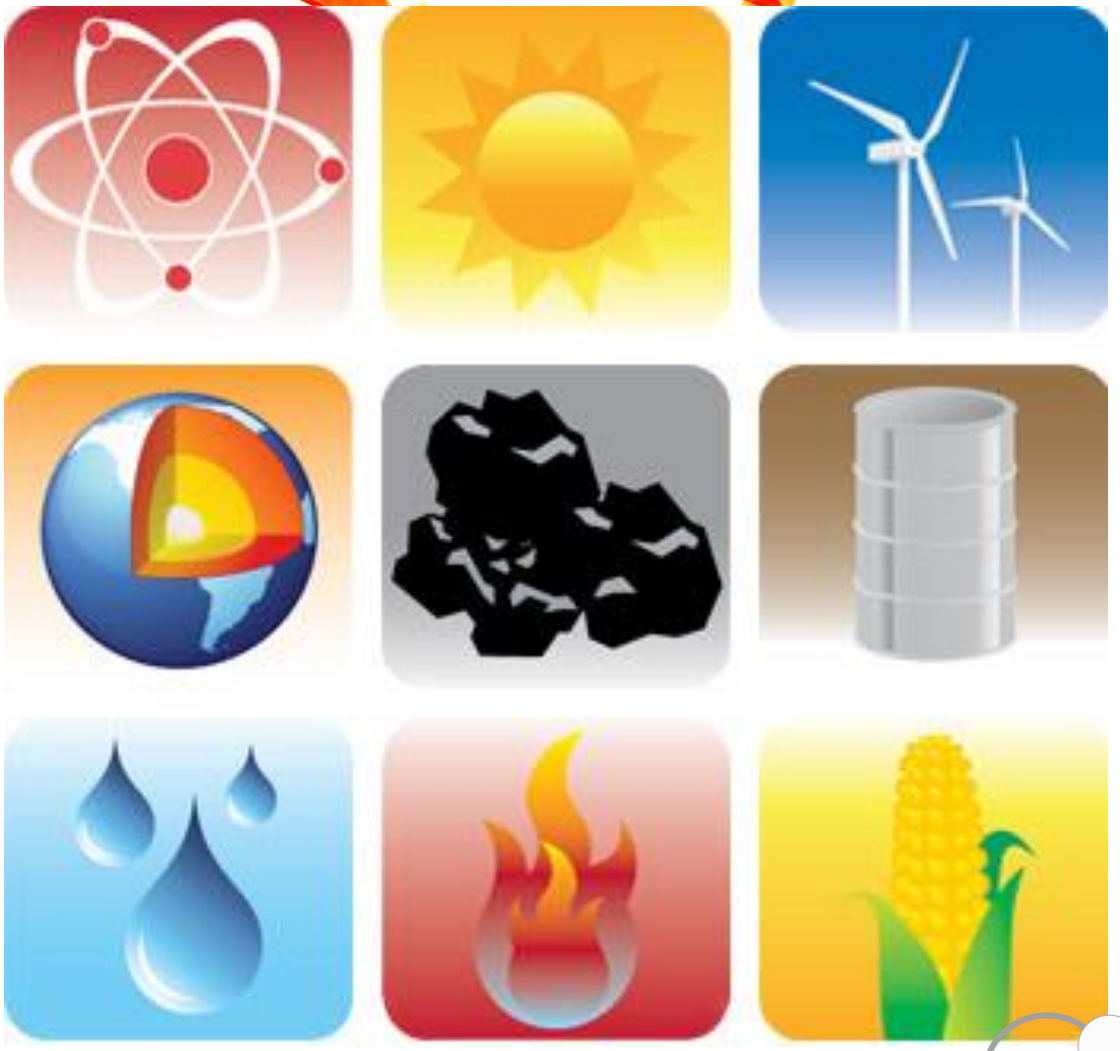
1a

Energy makes things happen

Energy is not a substance or an object that you can touch or hold, but substances and objects can **possess energy**

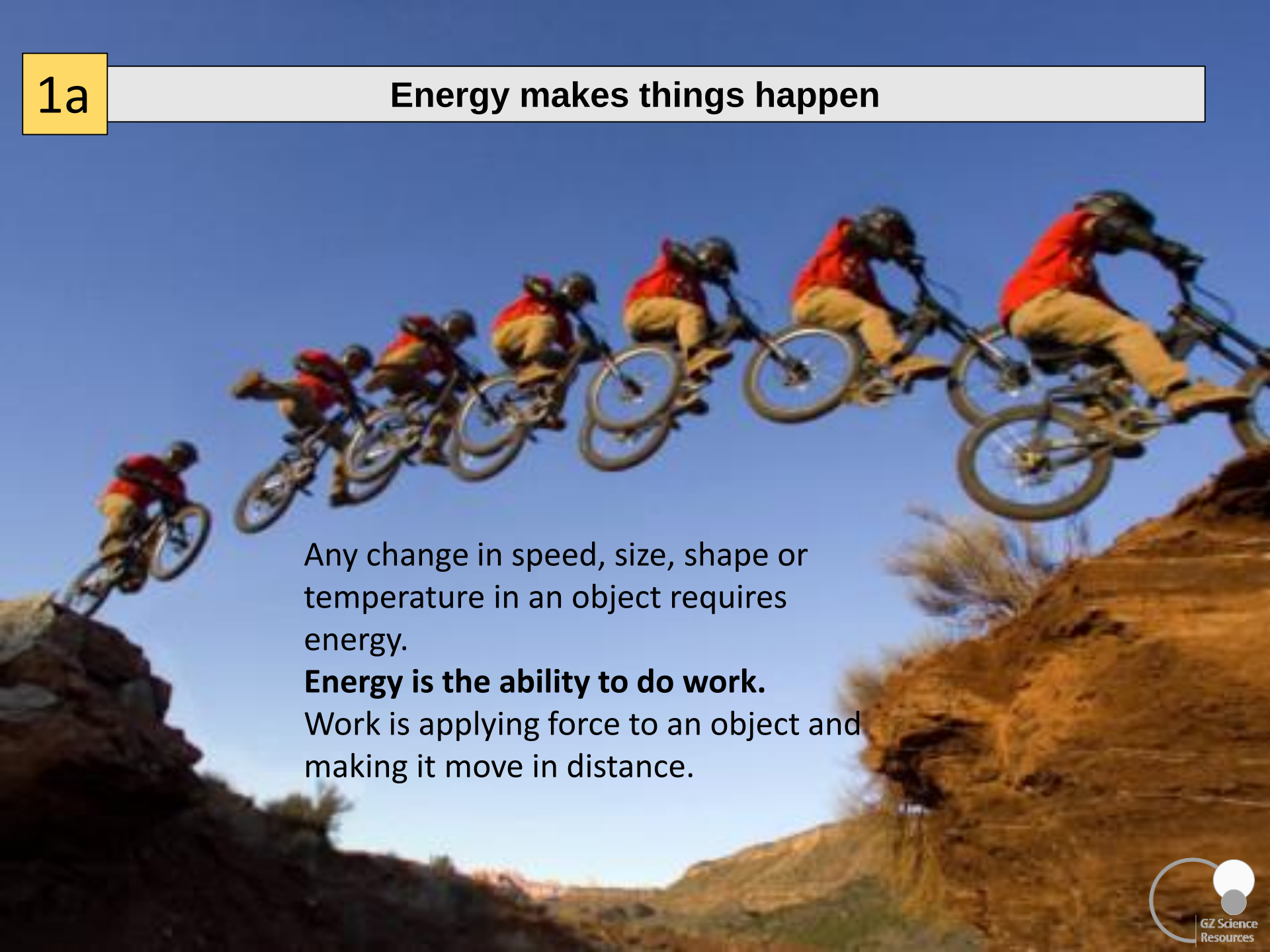
Energy is something that is required to make things change

Energy is needed to make objects move or change their state



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1a Energy makes things happen



Any change in speed, size, shape or temperature in an object requires energy.

Energy is the ability to do work.

Work is applying force to an object and making it move in distance.

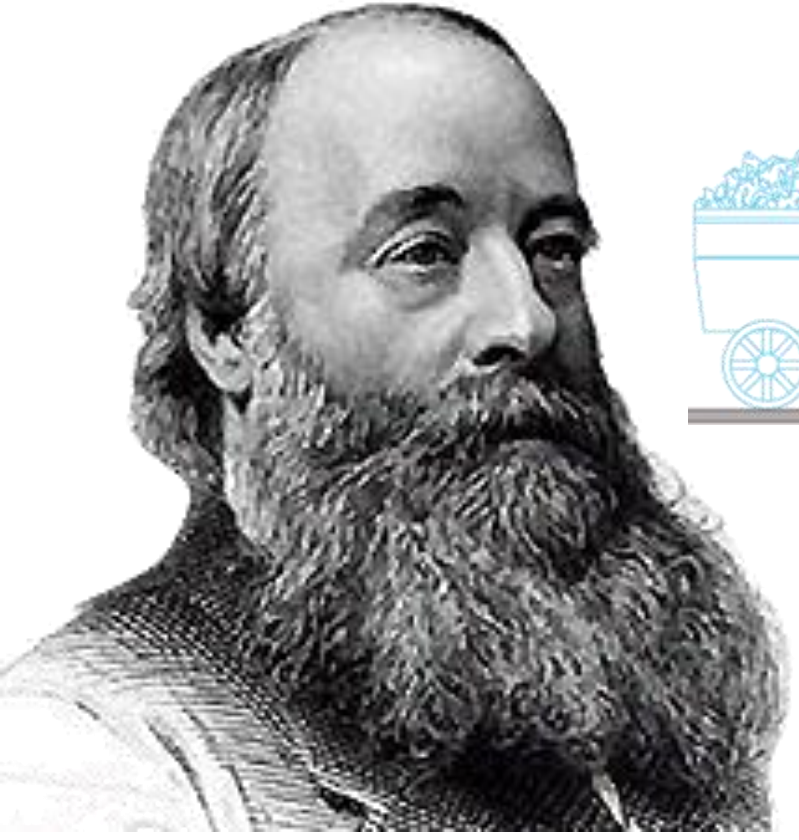
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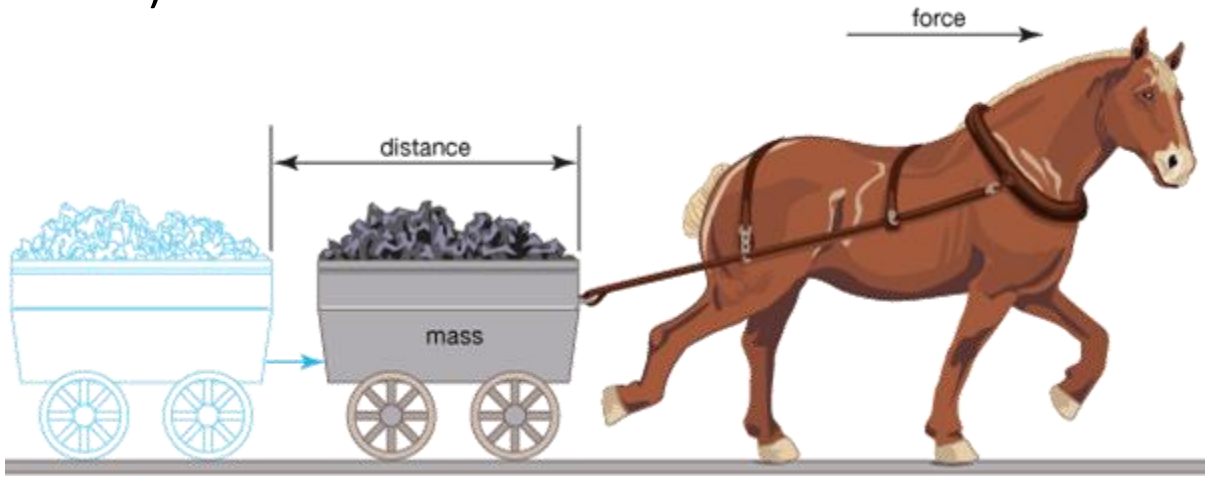
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Energy is measured in a unit called a joule (J).

A **joule**, unit of work or energy, is equal to the work done by a force of one **Newton** acting through **one metre**. This unit is named in honour of the English physicist James Prescott Joule.

W (work) = F (force) x d (distance)





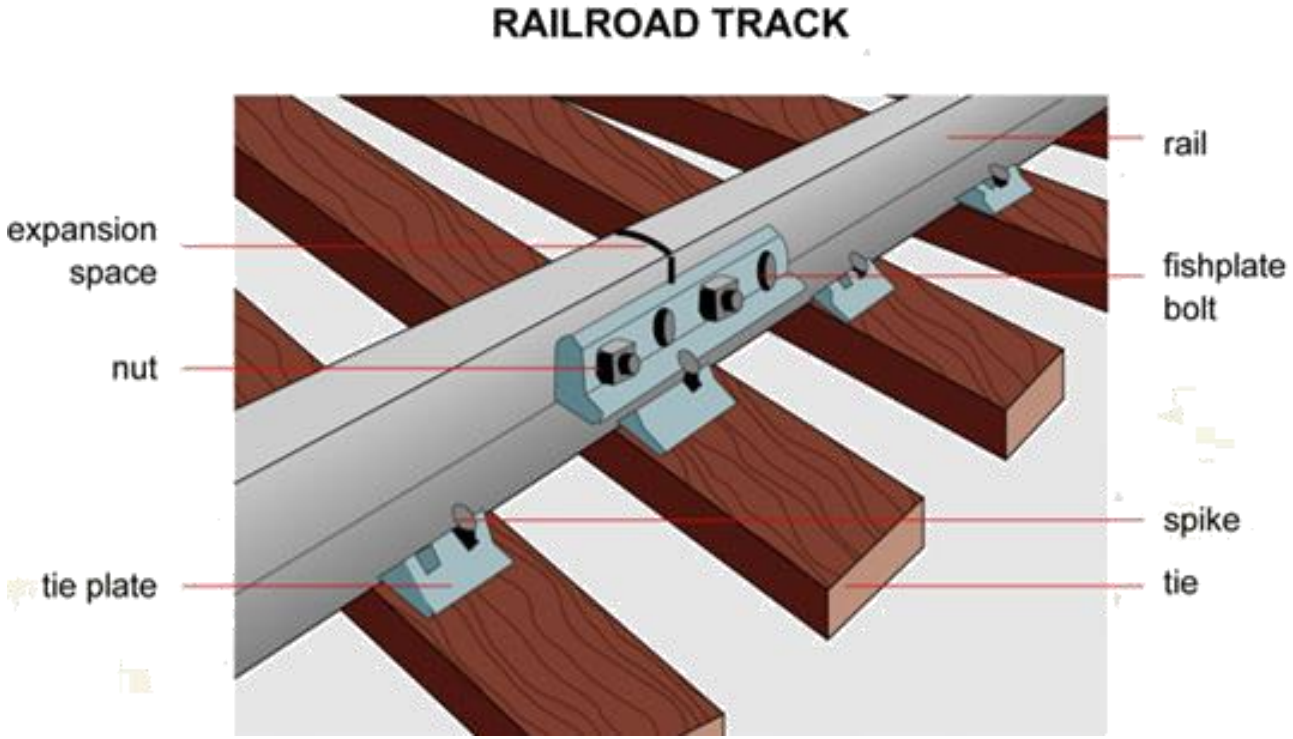
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4a/b Expansion and Contraction of matter

Matter **expands** or gets bigger when it is heated up. It **contracts** or gets smaller when it is cooled down.

Some types of matter, such as metals, conduct heat very well and therefore the effects of expansion and contraction are more obvious.

Structures such as bridges and railway tracks that are made out of metal have to be built with **gaps** so they don't warp out of shape when exposed to higher temperatures during summer.



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4a/b

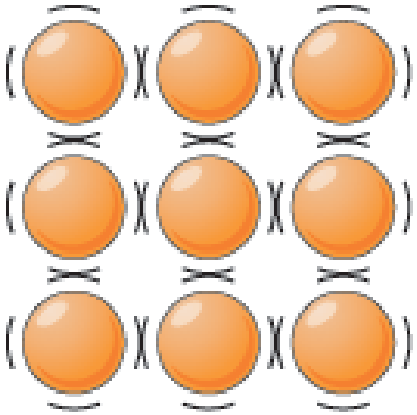
Expansion and Contraction of matter

When matter expands or contracts only the **space between** the particles changes: not the size of the particles.

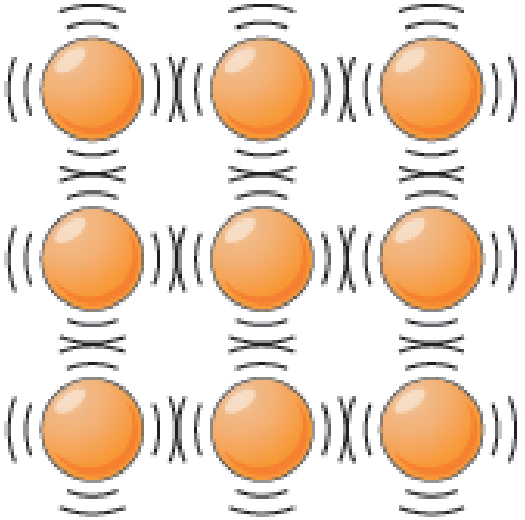
The particles in a **solid** vibrate more when it is heated, and take up more space.

The particles in a **liquid** move around each other more when it is heated, and take up more space.

The particles in a **gas** move more quickly in all directions when it is heated, and take up more space.



Cold



Hot

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4a/b Expansion, Contraction and Density in Gases

As matter expands the density (mass of matter in a given volume) will decrease making it "lighter".

In a gas the warmer particles that are moving faster due to their increased kinetic energy are colliding with each other more. This creates more space between the particles and therefore makes that area of gas less dense. The less dense gas will rise above the colder more dense. This is why hot air rises to the top of a ceiling in a heated house.



Heat expands the space between air particles and cause the balloon to inflate when heated



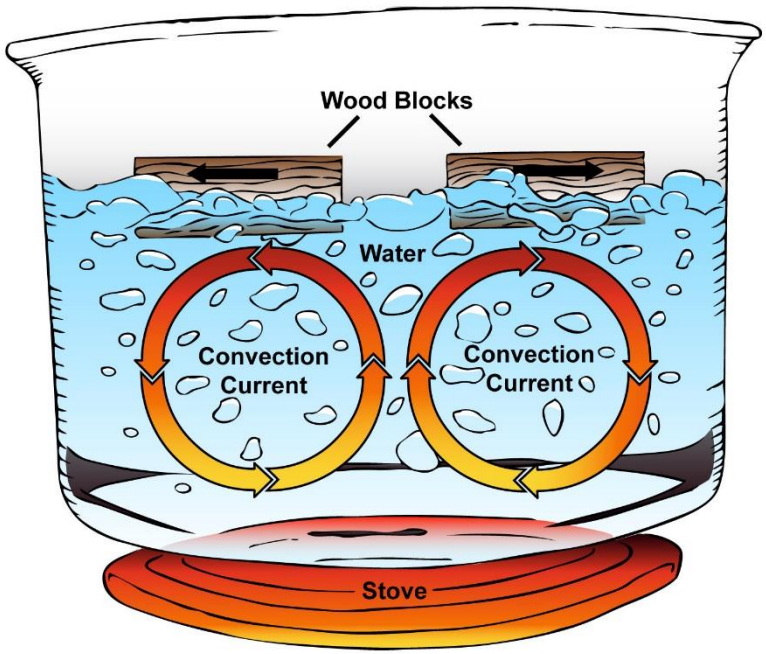
Heating water to create expanding steam powered many engines and began the industrial revolution

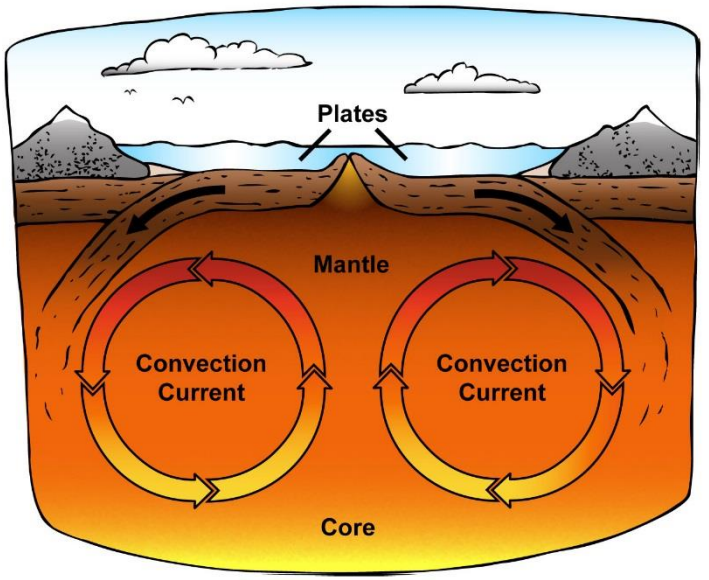
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4a/b Expansion, Contraction and Density in liquids

In liquids the hotter and expanded, less dense substance rises above cooler and denser liquids. This creates **convection currents** and causes the movement of particles containing heat energy.

Molten Magma under the earth's surface moves due to convection currents when the heat from the Earth's core causes it to expand. When the magma reaches the earth's surface under the crust it cools, contracts and gains in density, so it sinks again.





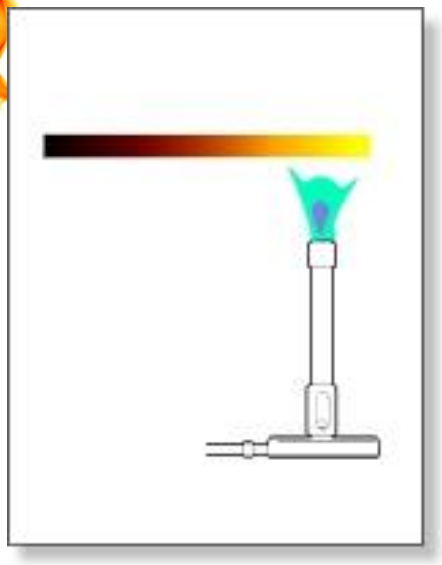
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Energy can be transferred by conduction, convection or radiation

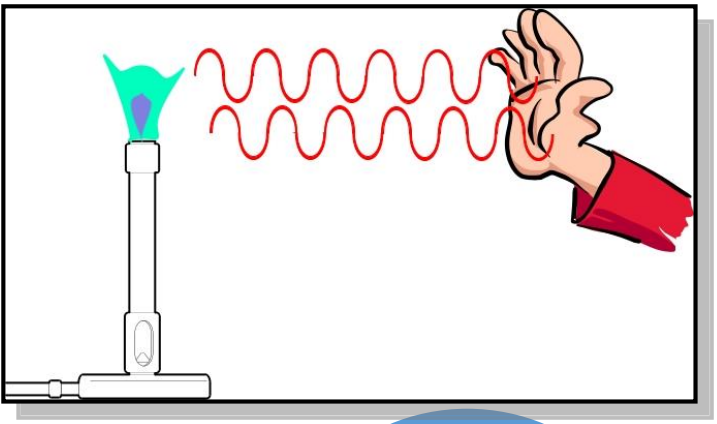
7a

Methods of heat transfer

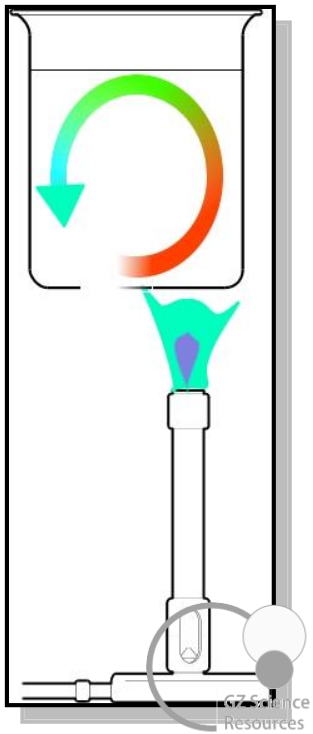
Conduction of heat along a solid object



Radiation of heat waves from hot objects

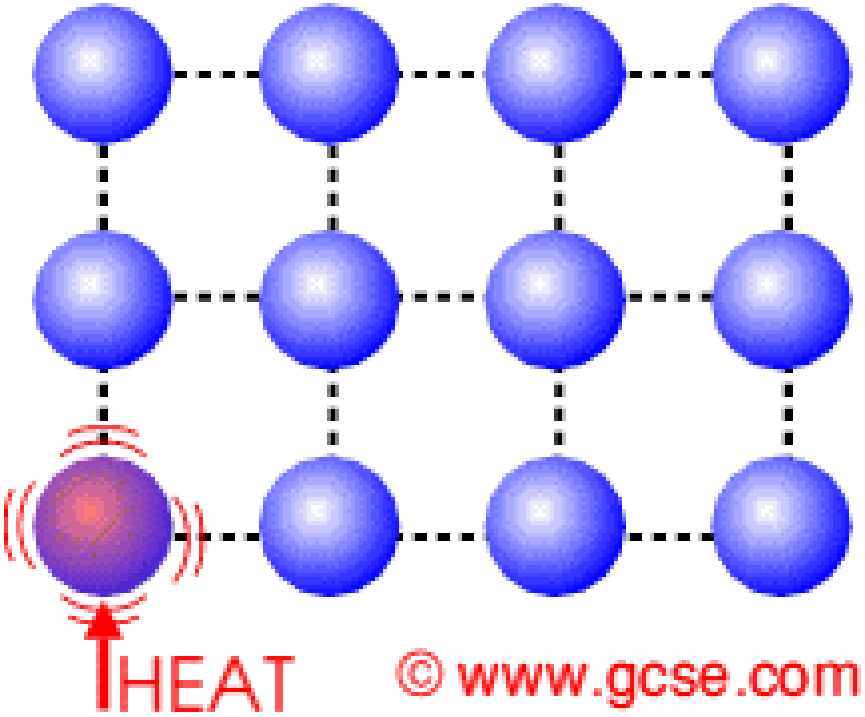


Convection currents in liquids or gases



7b Solids transfer heat energy by Conduction

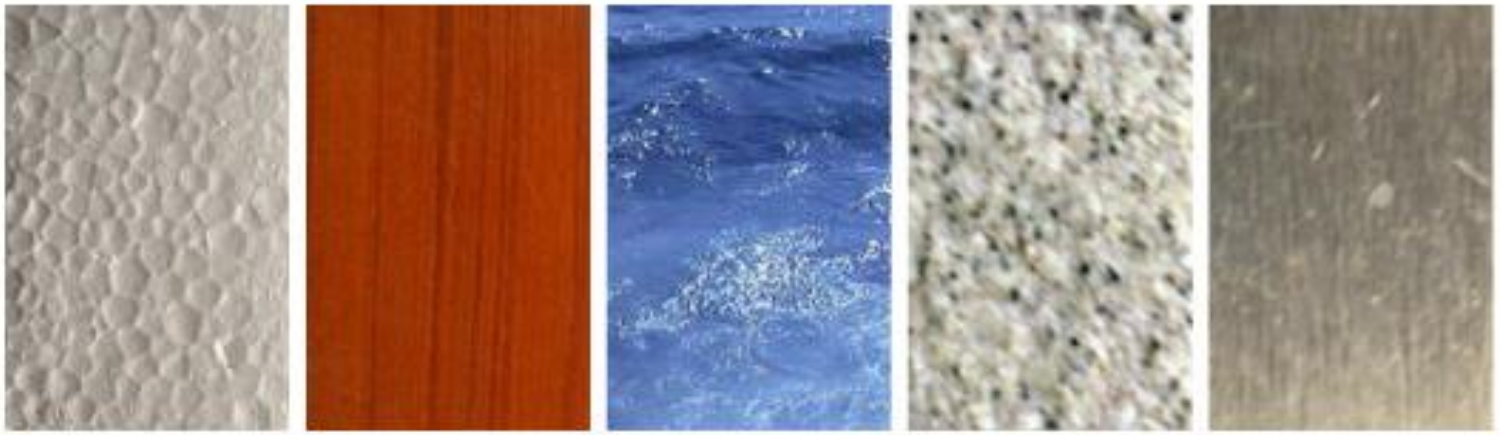
When heat is transferred by conduction – the atoms remain in fixed solid position – and the heat energy is transferred from one atom to another by being carried by moving electrons surrounding the atoms. The more heat energy an electron has the faster it moves. Some materials conduct better than others. They are called **conductors** and include metals which have electrons that are free to move. Those materials that cannot conduct heat are known as **insulators**. They have no free electrons to carry the heat energy.



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7c

Conductors and insulators



More insulating

Man made plastics are good insulators of heat because they have no free moving electrons or charged particles.

More conducting

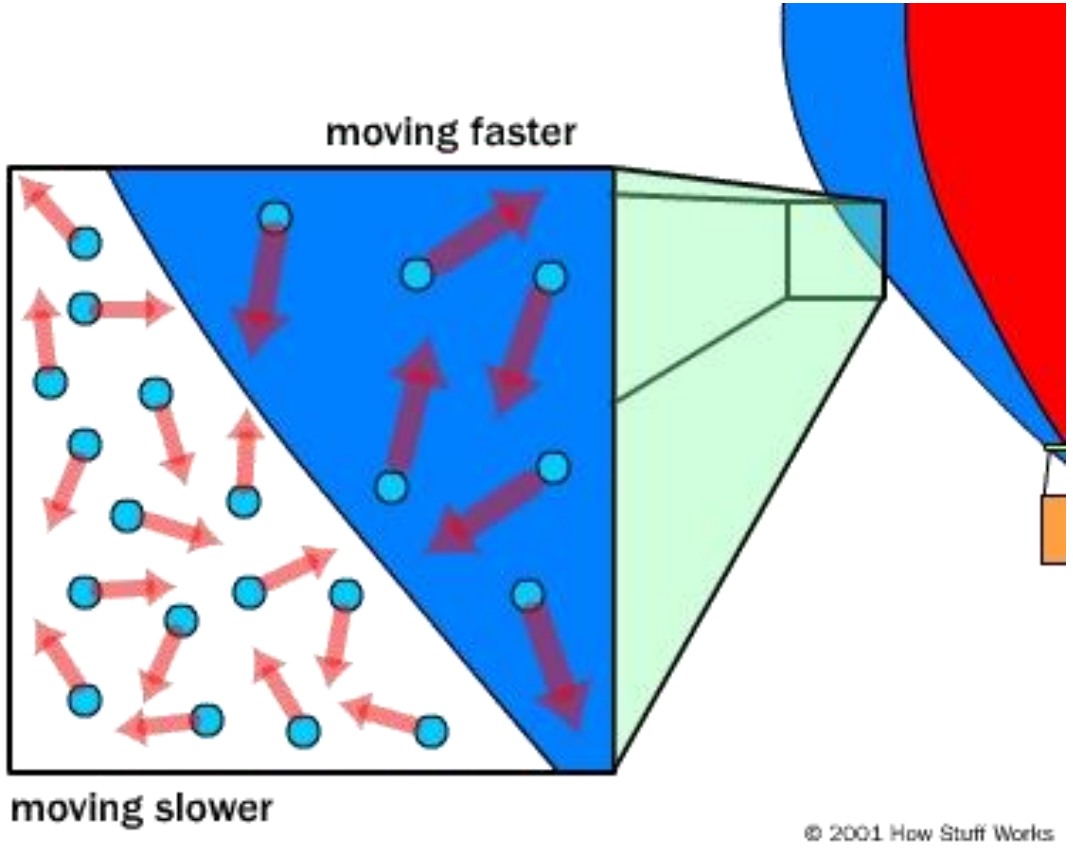
Metals and stone are good conductors of heat because they have free moving electrons to carry the heat.

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Gas and Liquids transfer Heat Energy by Convection

8b

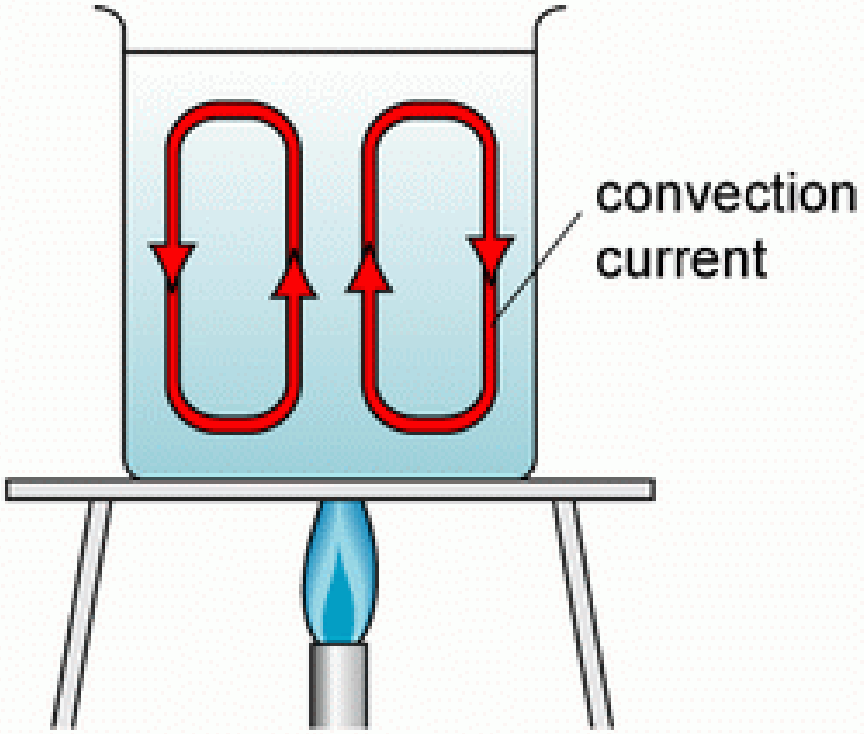
When you heat the air in the balloon your increase the kinetic energy of the air particles. The air particles absorb the heat energy and move faster. This makes them move faster. Air particles are pushed out of the balloon, as they take up more space.
A hot air balloon rises because it is filled with hot, less dense air and is surrounded by colder, more dense air.



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Convection Currents

8c

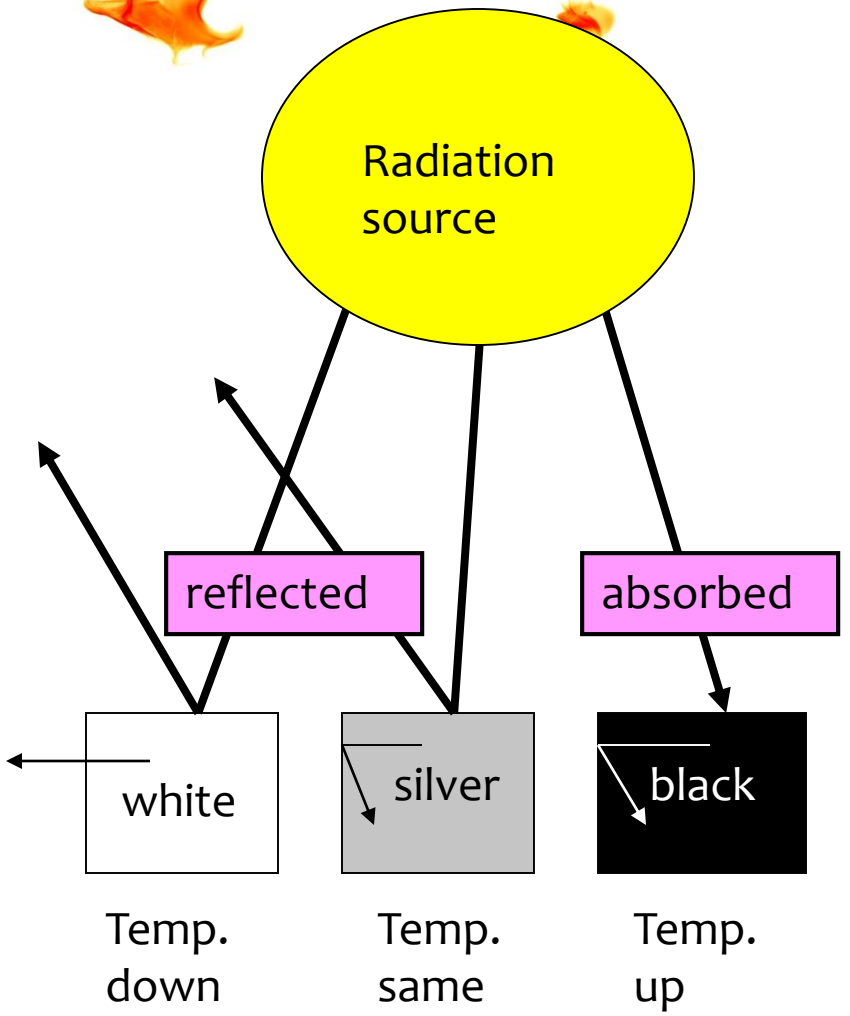


Convection occurs when free moving particles found in liquids and gases have heat energy added to them. The gas/liquid particles gains kinetic energy and as they collide into each other they push apart more. The hotter area of liquid or gas becomes less dense or lighter and rises away from the heat source. As the particles lose energy they slow down and become closer to each other which makes them denser and they fall back down. This movement creates **convection currents** where the particles circulate.

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9a Vacuums transfer heat energy by Radiation

Heat Energy can transferred in waves – it does not need atoms or electrons to travel. Our greatest source of radiation waves comes from the sun.
White and silver objects reflect radiation and very little heat energy is transferred to them. Black objects absorb radiation and a large proportion of the heat energy is transferred to them.



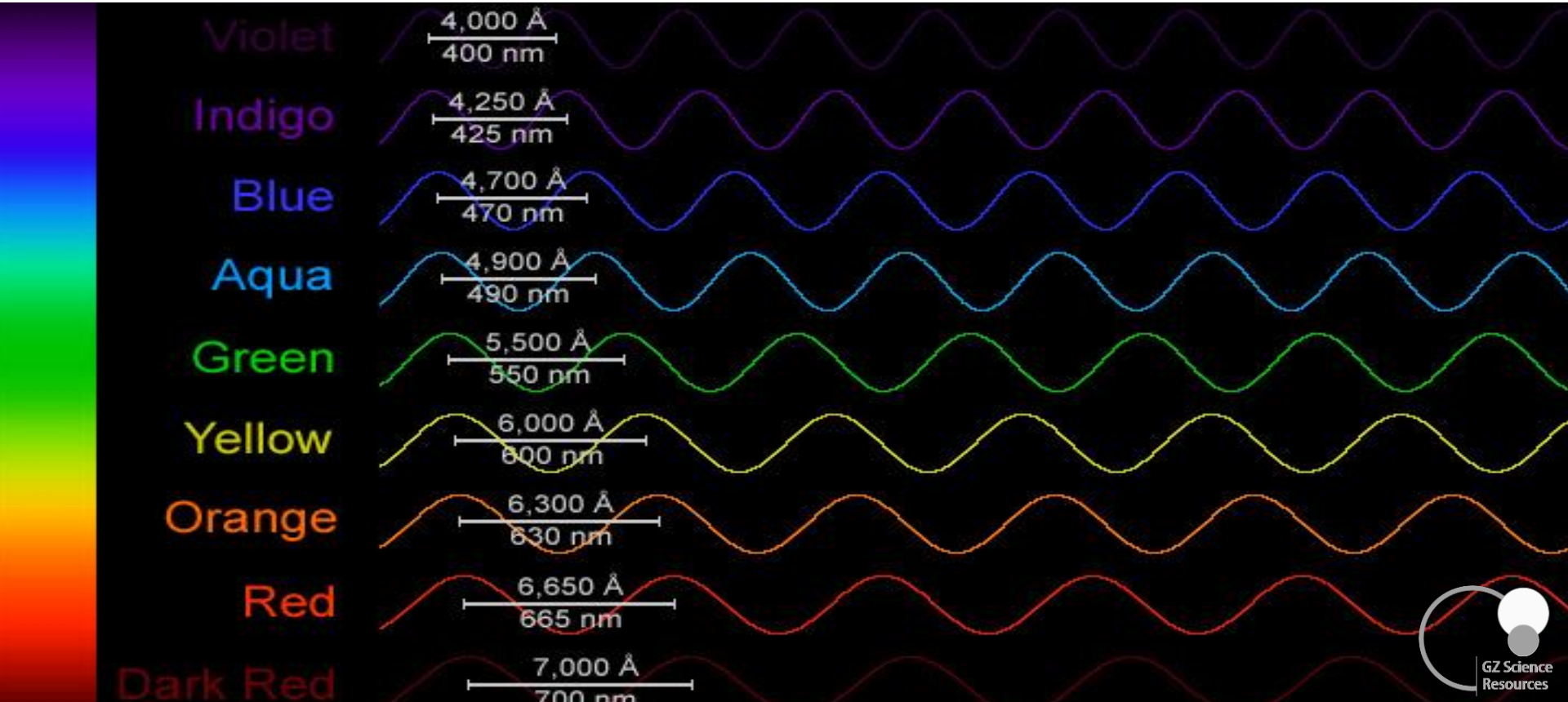
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9b Energy can be transferred as waves

Extra for experts

Light and other types of electromagnetic radiation from the sun and even further away stars travel through space in a vacuum – an area of very little or no atoms. Light does not need matter or a substance through which to travel.

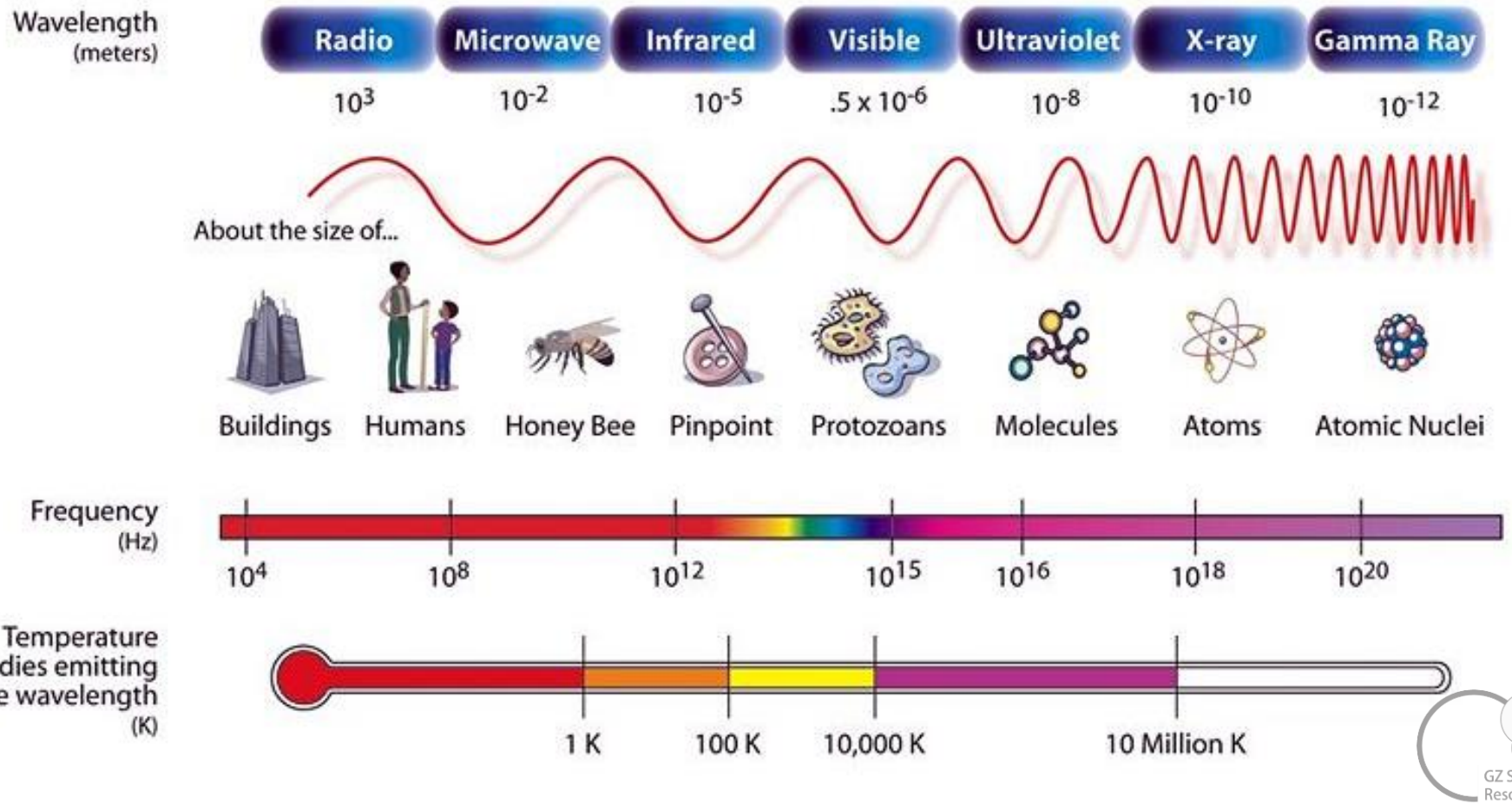
Each particular type of electromagnetic radiation, including each different colour of light, has a unique fixed length of wave, called the **wavelength** (λ), that it travels in.



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9b Heat energy belongs to the electromagnetic spectrum

Heat is a type of energy called electromagnetic (EM) radiation. There are other kinds of EM radiation such radio waves, microwaves, x-rays, etc. Heat travels as infrared waves. All of the types of EM radiation together are known as the **electromagnetic spectrum.**



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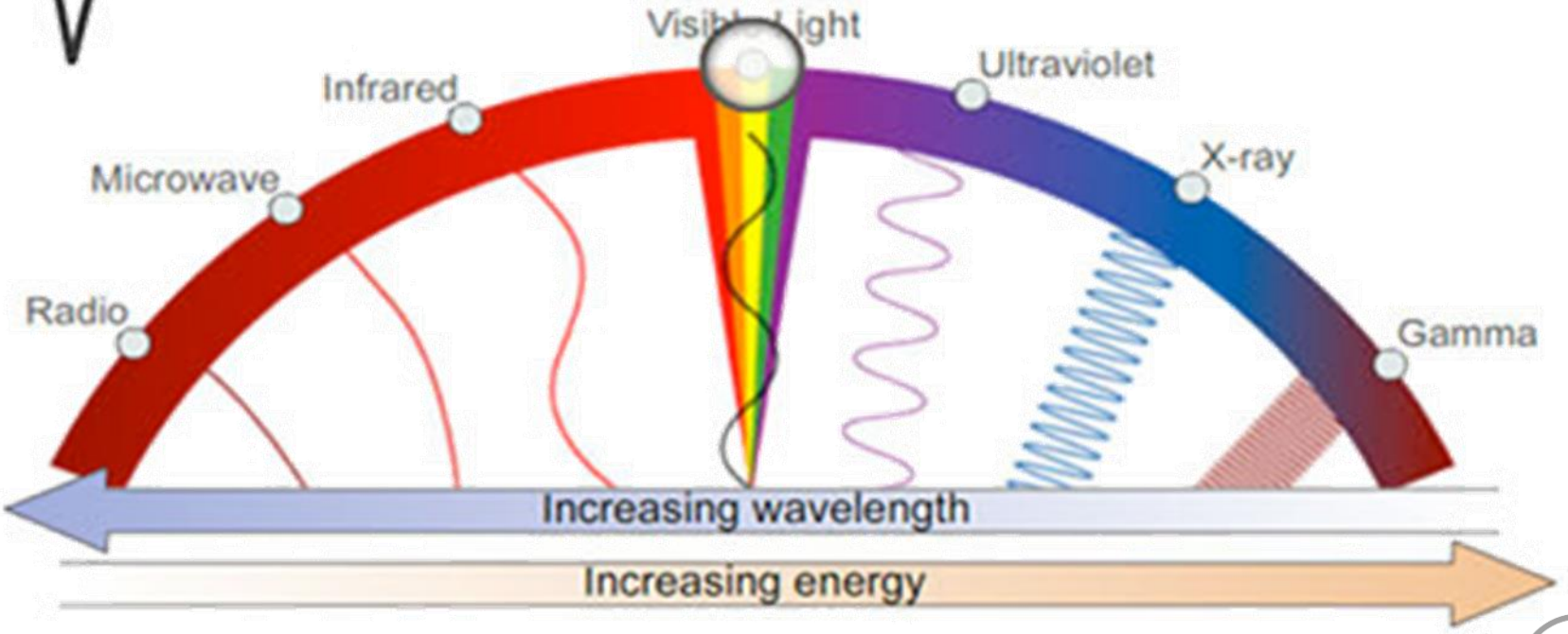
9b

The electromagnetic spectrum

Heat Energy

Heat energy, travelling as Infrared in wave form has a larger wavelength than visible light. A wave length is the distance from one wave to the next. Microwaves and radio waves have larger wavelengths than infrared waves. Some animals are able to "see" infrared and we can use special goggles to "see" heat.





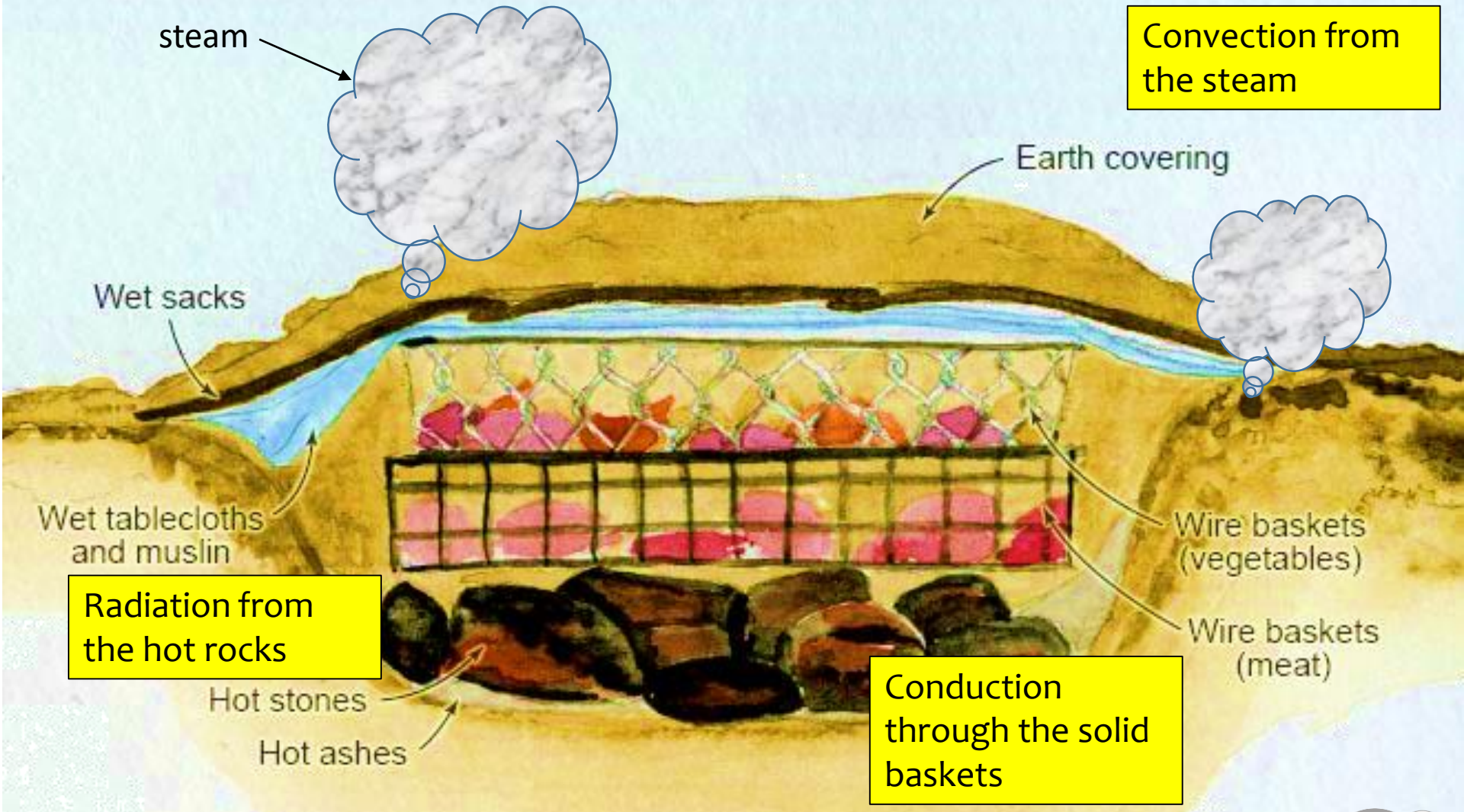
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9c The method of heat transfer is linked to the state of matter

Heat Transfer type	What does it travel in?	Does it require matter to travel?	Description	example
convection	Liquid And gas	yes	Heat energy contained by particles that move and carry it.	Hot air balloon rising when burner is going
conduction	solids	yes	Heat energy is passed from particle to particle.	A metal spoon heating up in a hot drink
Radiation	In a vacuum	no	Heat energy travelling in infra-red waves at the speed of light	Skin warming up while sitting in the sun

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9c Some situations transfer heat by many ways



In this Hangi there are many types of heat transfer occurring

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9c The method of heat transfer is linked to the state of matter



Conduction passes the heat energy from particle to particle and occurs in solids when the particles are fixed in place. This is like the rugby players passing the ball from one to another.



Convection requires particles to move around such as in liquid and gas and carry around the heat energy like a rugby player carrying the ball and running.



Radiation does not require particles to move the heat energy. This is like a rugby ball kicked which then moves by itself down the field without any players required.

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